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8824

Sections:

- TCP Server Code
- UDP Server Code
- Client Code
- 3 pages of screenshots of wireshark capturing at the end

tcp_server.py

```
import socket
import threading

def handle_connection(conn, addr):
    print(f"Client connected: {addr}")
    try:
        # Read until client closes
        buffer = []
        while True:
            chunk = conn.recv(1024)
            if not chunk:
                break
            buffer.append(chunk)

        message = b''.join(buffer).decode('utf-8').strip()
        result = process_message(message)
        conn.sendall(result.encode('utf-8'))
    except:
        pass
    finally:
        conn.close()
        print(f"Client disconnected: {addr}")

# Process message
def process_message(msg):
    if not msg:
        return ""

    cmd = msg[0]
    text = msg[1:]

    # A: sort descending
    if cmd == 'A':
        return "".join(sorted(text, reverse=True))
    # C: sort ascending
    elif cmd == 'C':
        return "".join(sorted(text))
    # D: uppercase
    elif cmd == 'D':
        return text.upper()
    # Unknown command, return as is
    else:
        return msg

# Main server function
def run_server():
    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    sock.bind(('127.0.0.1', 12345))
    sock.listen(5)
    print("TCP Server started on port 12345")

    while True:
        client_conn, client_addr = sock.accept()
        # Use threading for multiple clients
        t = threading.Thread(target=handle_connection, args=(client_conn,
client_addr))
        t.start()

if __name__ == "__main__":
```

```
run_server()
```

upd_server.py

```
import socket

def process_message(msg):
    if not msg:
        return ""

    cmd = msg[0]
    text = msg[1:]

    if cmd == 'A':
        return "".join(sorted(text, reverse=True))
    elif cmd == 'C':
        return "".join(sorted(text))
    elif cmd == 'D':
        return text.upper()
    else:
        return msg

# UDP server
def run_server():
    sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    sock.bind(('127.0.0.1', 12346))
    print("UDP Server started on port 12346")

    while True:
        data, client_addr = sock.recvfrom(1024)
        message = data.decode('utf-8').strip()
        print(f"Received from {client_addr}: {message}")

        result = process_message(message)
        sock.sendto(result.encode('utf-8'), client_addr)

if __name__ == "__main__":
    run_server()
```

Client.py

```
import socket

# Connect and send message to server
def send_message(protocol, port):
    server_addr = ('127.0.0.1', port)
    message = input(f"Enter message for {protocol}: ")

    if protocol == 'TCP':
        # TCP connection
        sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```

    sock.connect(server_addr)
    sock.sendall(message.encode('utf-8'))
    sock.shutdown(socket.SHUT_WR)
    response = sock.recv(1024)
    print(f"Server reply: {response.decode('utf-8')}")
    sock.close()
else:
    # UDP connection
    sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    sock.sendto(message.encode('utf-8'), server_addr)
    response, _ = sock.recvfrom(1024)
    print(f"Server reply: {response.decode('utf-8')}")
    sock.close()

if __name__ == "__main__":
    choice = input("Choose protocol (TCP/UDP): ").upper()
    port_num = 12345 if choice == 'TCP' else 12346
    send_message(choice, port_num)

```

tcp.port == 12345 udp.port == 12346						
No.	Time	Source	Destination	Protocol	Length	Info
6	22.350070	127.0.0.1	127.0.0.1	TCP	56	54786 → 12345 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=3472984239 TSecr=545470553
7	22.350079	127.0.0.1	127.0.0.1	TCP	62	54786 → 12345 [PSH, ACK] Seq=1 Ack=1 Win=408320 Len=6 TSval=3472984239 TSecr=545470553
8	22.350083	127.0.0.1	127.0.0.1	TCP	56	54786 → 12345 [FIN, ACK] Seq=7 Ack=1 Win=408320 Len=0 TSval=3472984239 TSecr=545470553
9	22.350091	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 12345 → 54786 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=545470553 TSecr=3472984239
10	22.350105	127.0.0.1	127.0.0.1	TCP	56	12345 → 54786 [ACK] Seq=1 Ack=7 Win=408320 Len=0 TSval=545470553 TSecr=3472984239
11	22.350113	127.0.0.1	127.0.0.1	TCP	56	12345 → 54786 [ACK] Seq=1 Ack=8 Win=408320 Len=0 TSval=545470553 TSecr=3472984239
12	22.350292	127.0.0.1	127.0.0.1	TCP	61	12345 → 54786 [PSH, ACK] Seq=1 Ack=8 Win=408320 Len=5 TSval=545470553 TSecr=3472984239
13	22.350300	127.0.0.1	127.0.0.1	TCP	56	12345 → 54786 [FIN, ACK] Seq=6 Ack=8 Win=408320 Len=0 TSval=545470553 TSecr=3472984239
14	22.350335	127.0.0.1	127.0.0.1	TCP	56	54786 → 12345 [ACK] Seq=8 Ack=6 Win=408320 Len=0 TSval=3472984239 TSecr=545470553
15	22.350371	127.0.0.1	127.0.0.1	TCP	56	54786 → 12345 [ACK] Seq=8 Ack=7 Win=408320 Len=0 TSval=3472984239 TSecr=545470553
16	22.855553	127.0.0.1	127.0.0.1	TCP	68	54787 → 12345 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=198391141 TSecr=0 SACK_PERM
17	22.855755	127.0.0.1	127.0.0.1	TCP	68	12345 → 54787 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=3262772029 TSecr=198391141 SACK_P...
18	22.855791	127.0.0.1	127.0.0.1	TCP	56	54787 → 12345 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=198391141 TSecr=3262772029
19	22.855815	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 12345 → 54787 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=3262772029 TSecr=198391141
20	22.855827	127.0.0.1	127.0.0.1	TCP	62	54787 → 12345 [PSH, ACK] Seq=1 Ack=1 Win=408320 Len=6 TSval=198391141 TSecr=3262772029
21	22.855846	127.0.0.1	127.0.0.1	TCP	56	54787 → 12345 [FIN, ACK] Seq=7 Ack=1 Win=408320 Len=0 TSval=198391141 TSecr=3262772029
22	22.855851	127.0.0.1	127.0.0.1	TCP	56	12345 → 54787 [ACK] Seq=1 Ack=7 Win=408320 Len=0 TSval=3262772029 TSecr=198391141
23	22.855863	127.0.0.1	127.0.0.1	TCP	56	12345 → 54787 [ACK] Seq=1 Ack=8 Win=408320 Len=0 TSval=3262772029 TSecr=198391141
24	22.856289	127.0.0.1	127.0.0.1	TCP	61	12345 → 54787 [PSH, ACK] Seq=1 Ack=8 Win=408320 Len=5 TSval=3262772029 TSecr=198391141
25	22.856308	127.0.0.1	127.0.0.1	TCP	56	12345 → 54787 [FIN, ACK] Seq=6 Ack=8 Win=408320 Len=0 TSval=3262772029 TSecr=198391141
26	22.856326	127.0.0.1	127.0.0.1	TCP	56	54787 → 12345 [ACK] Seq=8 Ack=6 Win=408320 Len=0 TSval=198391141 TSecr=3262772029
27	22.856361	127.0.0.1	127.0.0.1	TCP	56	54787 → 12345 [ACK] Seq=8 Ack=7 Win=408320 Len=0 TSval=198391141 TSecr=3262772029
28	23.358700	127.0.0.1	127.0.0.1	TCP	68	54788 → 12345 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=142908833 TSecr=0 SACK_PERM
29	23.358886	127.0.0.1	127.0.0.1	TCP	68	12345 → 54788 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=2402087604 TSecr=142908833 SACK_P...
30	23.358938	127.0.0.1	127.0.0.1	TCP	56	54788 → 12345 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=142908833 TSecr=2402087604
31	23.358956	127.0.0.1	127.0.0.1	TCP	62	54788 → 12345 [PSH, ACK] Seq=1 Ack=1 Win=408320 Len=6 TSval=142908833 TSecr=2402087604
32	23.358968	127.0.0.1	127.0.0.1	TCP	56	54788 → 12345 [FIN, ACK] Seq=7 Ack=1 Win=408320 Len=0 TSval=142908833 TSecr=2402087604
33	23.358972	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 12345 → 54788 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=2402087604 TSecr=142908833
34	23.358993	127.0.0.1	127.0.0.1	TCP	56	12345 → 54788 [ACK] Seq=1 Ack=7 Win=408320 Len=0 TSval=2402087604 TSecr=142908833
35	23.359002	127.0.0.1	127.0.0.1	TCP	56	12345 → 54788 [ACK] Seq=1 Ack=8 Win=408320 Len=0 TSval=2402087604 TSecr=142908833
36	23.359255	127.0.0.1	127.0.0.1	TCP	61	12345 → 54788 [PSH, ACK] Seq=1 Ack=8 Win=408320 Len=5 TSval=2402087604 TSecr=142908833
37	23.359265	127.0.0.1	127.0.0.1	TCP	56	12345 → 54788 [FIN, ACK] Seq=6 Ack=8 Win=408320 Len=0 TSval=2402087604 TSecr=142908833
38	23.359309	127.0.0.1	127.0.0.1	TCP	56	54788 → 12345 [ACK] Seq=8 Ack=6 Win=408320 Len=0 TSval=142908833 TSecr=2402087604
39	23.359353	127.0.0.1	127.0.0.1	TCP	56	54788 → 12345 [ACK] Seq=8 Ack=7 Win=408320 Len=0 TSval=142908833 TSecr=2402087604
40	23.861715	127.0.0.1	127.0.0.1	UDP	38	62289 → 12346 Len=6
41	23.862280	127.0.0.1	127.0.0.1	UDP	37	12346 → 62289 Len=5
42	24.363508	127.0.0.1	127.0.0.1	UDP	38	58733 → 12346 Len=6
43	24.363766	127.0.0.1	127.0.0.1	UDP	37	12346 → 58733 Len=5
44	24.869271	127.0.0.1	127.0.0.1	UDP	37	50407 → 12346 Len=5
45	24.869519	127.0.0.1	127.0.0.1	UDP	36	12346 → 50407 Len=4

Wireshark · Follow TCP Stream (tcp.stream eq 2) · Loopback: lo0

Dhello
HELLO

1 client pkt, 1 server pkt, 1 turn.

Entire conversation (11 bytes) Show as ASCII No delta times Stream 2

Find: ☐ Case sensitive Find Next

Help

Filter Out This Stream

Print

Save as...

Back

Close

Azyxwv
zyxwv

1 **client** pkt, 1 **server** pkt, 1 turn.

Entire conversation (11 bytes)

Show as

ASCII

No delta times

Stream

3

Find:

☐ Case sensitive

Find Next

Help

Filter Out This Stream

Print

Save as...

Back

Close